

Two-Wheeler Resale Market Analysis in India

1. Project Overview and Objective

Project Overview:

This project is about the two-wheeler resale market in India. The main focus of the project is to understand the resale prices of bikes and the different factors that affect their prices. The data is cleaned and prepared using **Python** with libraries like **Pandas** and **NumPy**. **Seaborn** and **Matplotlib** are used to create charts and understand the data better. Descriptive Analysis is used to find the main trends and patterns in the data, while Diagnostic Analysis is used to understand the reasons behind those patterns. Finally, the analysed data is presented in an interactive **Power BI** dashboard.

Objective:

The main objective of this project is to analyse the Indian two-wheeler resale market and understand the factors that affect resale prices. The project aims to identify useful trends and patterns based on factors such as location, fuel type, brand, model, owner type, insurance status, and seller type. It also aims to compare the original price and resale price to understand the depreciation of two-wheelers. The final insights are presented through a Power BI dashboard to make the analysis easy to understand.

2. Data Sources

- **Source Description and Timeline:** Hugging Face and 2021-2024.
 - **Domain:** Automobile Industry-Two-Wheeler Resale Market
-

3. Problem Statement

- Identify state-wise and city-tier-wise trends in the two-wheeler resale market in India.
 - Analyze the effect of fuel type (Petrol, Electric, Hybrid) on resale value.
 - Find the brands and models that retain better resale value.
 - Understand how owner type, insurance status, and seller type influence resale price.
 - Compare original price and resale price to study the depreciation pattern from 2021 to 2024.
-

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
State	Categorical	Indian state where the bike was sold

Avg Daily Distance (km)	Numerical (Float)	Average distance the bike is driven daily
Brand	Categorical	Manufacturer of the bike
Model	Categorical	Specific bike model
Price (INR)	Numerical (Float)	Showroom price of the bike
Year of Manufacture	Numerical (Integer)	Year the bike was manufactured
Engine Capacity (cc)	Numerical (Integer)	Engine capacity in cubic centimeters
Fuel Type	Categorical	Petrol, Electric, or Hybrid
Mileage (km/1)	Numerical (Float)	Fuel efficiency of the bike
Owner Type	Categorical	First, Second, or Third owner
Registration Year	Numerical (Integer)	Year the bike was registered
Insurance Status	Categorical	Whether insurance is active or expired
Seller Type	Categorical	Individual or Dealer
Resale Price (INR)	Numerical (Float)	Current resale value of the bike
City Tier	Categorical	Tier classification of the city (Tier 1/2/3)

5. Tools & Technologies

Python

- **Pandas** - Data Cleaning and Data Transformation
- **Seaborn** - Data Visualization and Exploratory Analysis

Power BI

- **Data Visualization** – Creating charts and graphs to present the analysis
 - **Interactive Dashboard** – Creating an interactive dashboard to display key insights
 - **DAX Measures** – Creating measures for KPI cards and required calculations
-

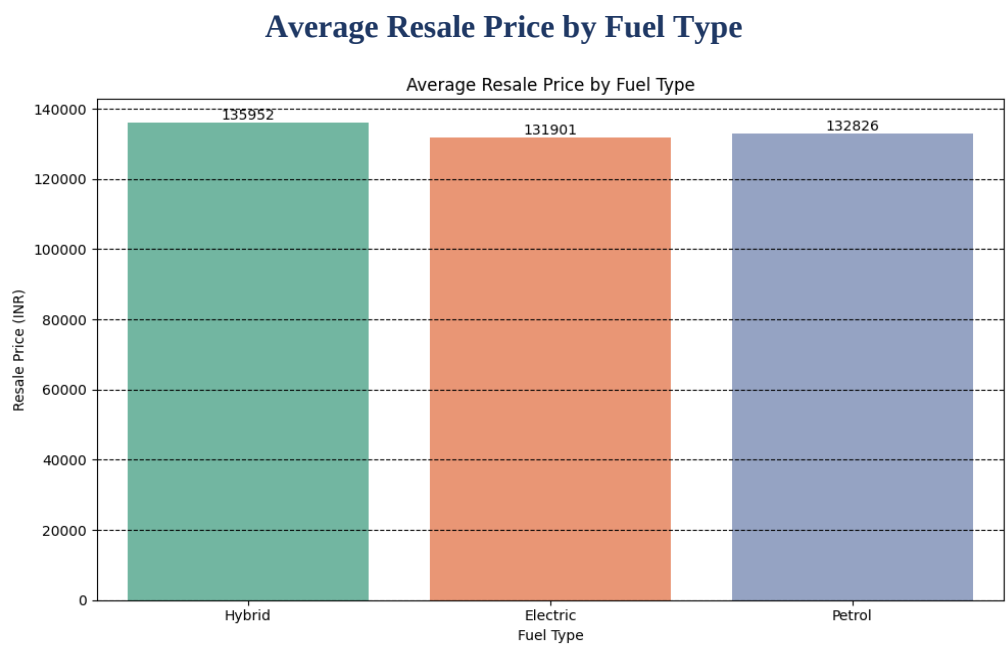
6. Data Pre-Processing (Python / Pandas)

Tasks Performed:

- **Handling Missing Values:** Checked the dataset for missing values and handled them using suitable methods such as mean, median, and mode. For categorical columns, the mode value was used where required. For example, missing values in Insurance Status were filled using the mode.
 - **Removing Duplicate Records:** Checked the dataset for duplicate rows and removed the duplicate records using the Pandas `drop_duplicates()` function.
 - **Data Type Conversion:** Converted columns into suitable data types for further analysis. Price values were converted from text format to numeric format, and year-related columns were converted into integer format.
 - **Text Cleaning and Standardization:** Removed unwanted spaces and standardized text values using Pandas string functions such as `str.strip()` and `str.title()`.
 - **Category Standardization:** Standardized categorical values such as Owner Type so that similar values were represented in the same format.
 - **Selecting Numerical Columns:** Selected the numerical columns from the dataset using Pandas. The numerical columns included Average Daily Distance, Price, Year of Manufacture, Engine Capacity, Mileage, Registration Year, and Resale Price.
 - **Skewness Check:** After handling the missing values, checked the skewness of the numerical columns. This helped to understand whether the data was positively or negatively skewed.
 - **Outlier Detection using IQR Method:** Checked for outliers in the numerical columns using the IQR (Interquartile Range) method. Q1, Q3 and IQR values were used to identify the lower and upper limits for detecting outliers.
 - **Outlier Handling:** The identified outlier values were handled using the median value. This helped to reduce the effect of extreme values while keeping the data available for further analysis.
 - **Final Data Validation:** After completing the cleaning process, checked the dataset again for missing values, duplicate records and data consistency before using the cleaned data for visualization and Power BI dashboard creation.
-

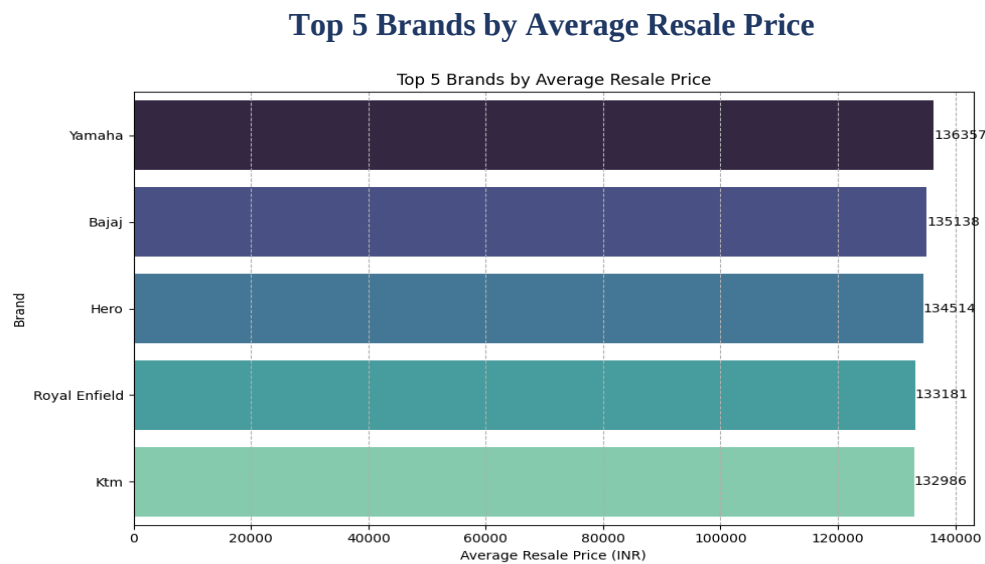
7. Data Visualization (Python)

Chart: Bar Plot



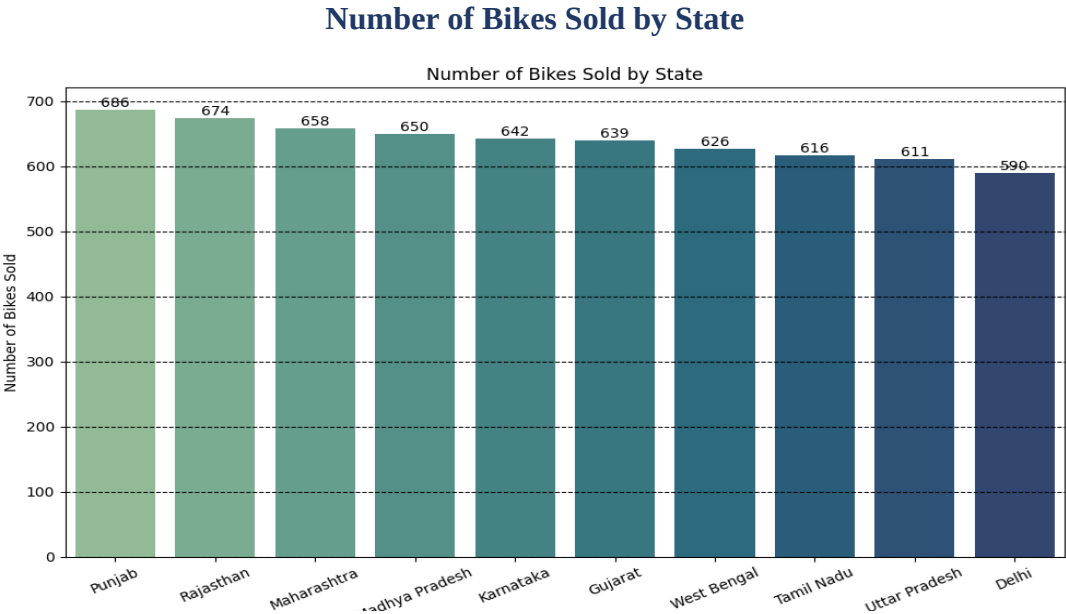
- **Interpretation:** Bar plot is used to easily compare the average resale price between different fuel types.
- Fuel Type is used on the X-axis and Resale Price (INR) on the Y-axis. Set2 palette and value labels are used to make the chart easy to understand.
- Hybrid has the highest average resale price (₹1,35,952), followed by Petrol (₹1,32,826) and Electric (₹1,31,901).
- The difference between the three fuel types is small, so the resale price is almost similar across fuel types, with Hybrid slightly higher.

Chart: Horizontal Bar Plot



- **Interpretation:** Horizontal bar plot is used to make the comparison of average resale prices between the top 5 brands easy to understand.
 - Brand is used on the Y-axis and Average Resale Price (INR) on the X-axis. mako palette, value labels, and horizontal grid lines are used for better readability.
 - Yamaha has the highest average resale price among the top 5 brands, followed by Bajaj and the other brands.
 - The chart helps us quickly identify the brands with higher average resale prices.
-

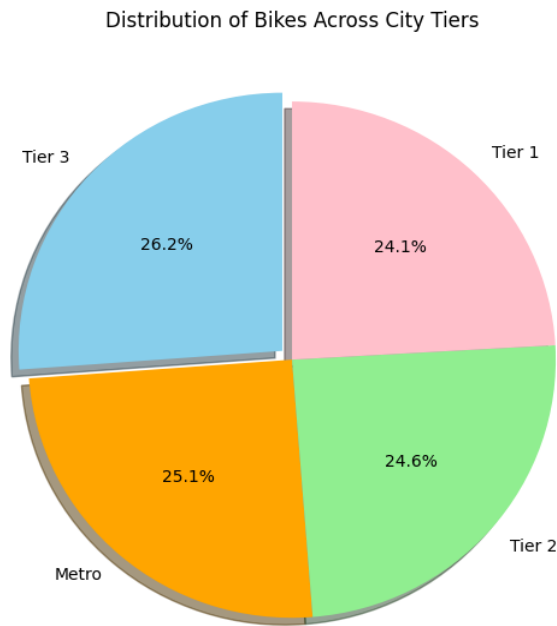
Chart: Bar Plot



- **Interpretation:** Bar plot is used to easily compare the number of bikes sold across different states.
 - State is used on the X-axis and Number of Bikes Sold on the Y-axis. crest palette, value labels, grid lines, and rotated labels are used for better readability.
 - Punjab has the highest number of bikes sold (686), followed by Rajasthan (674) and Maharashtra (658).
 - Delhi has the lowest sales among the selected states (590). Overall, the difference in sales between the states is noticeable but not very large.
-

Chart: Pie Plot

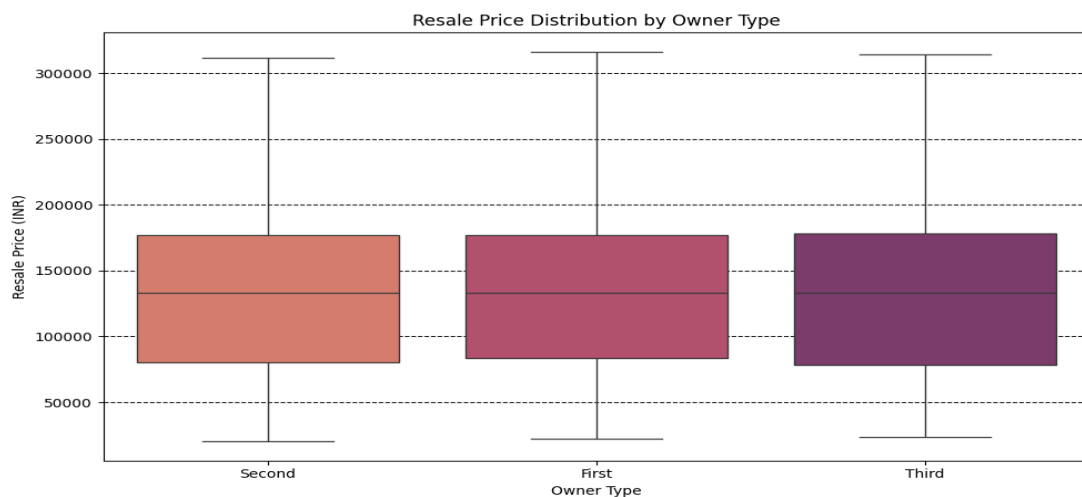
Distribution of Bikes Across City Tiers



- **Interpretation:** Pie chart is used to easily understand the distribution of bikes across different city tiers.
- City Tier and count are used to show the share of bikes, with percentage labels, custom colors, startangle, explode, and shadow used for better presentation.
- Tier 3 has the highest share of bikes (26.2%), followed by Metro (25.1%), Tier 2 (24.6%), and Tier 1 (24.1%).
- The distribution is almost evenly divided across the four city tiers, so no single tier has a very large difference compared with the others.

Chart: Box Plot

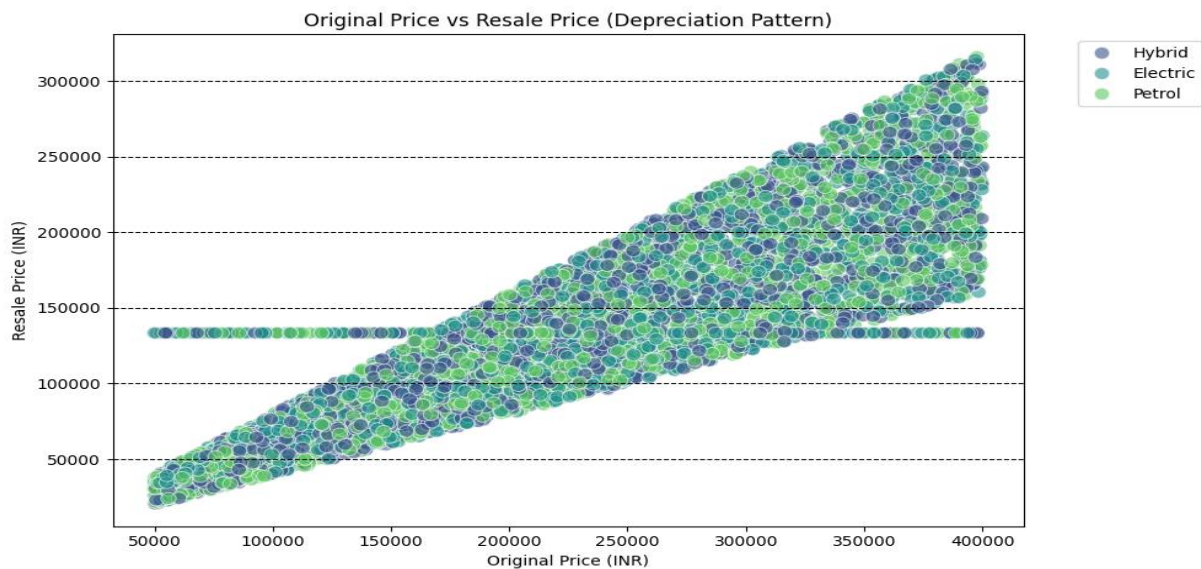
Resale Price Distribution by Owner Type



- **Interpretation:** Box plot is used to compare resale price distribution across owner types. Owner Type is on the X-axis and Resale Price is on the Y-axis, with flare palette and horizontal grid lines.
 - The median resale price is almost similar for First, Second, and Third owner types.
 - The chart shows that resale prices have a wide range within all three owner types, while the overall distributions are quite similar.
-

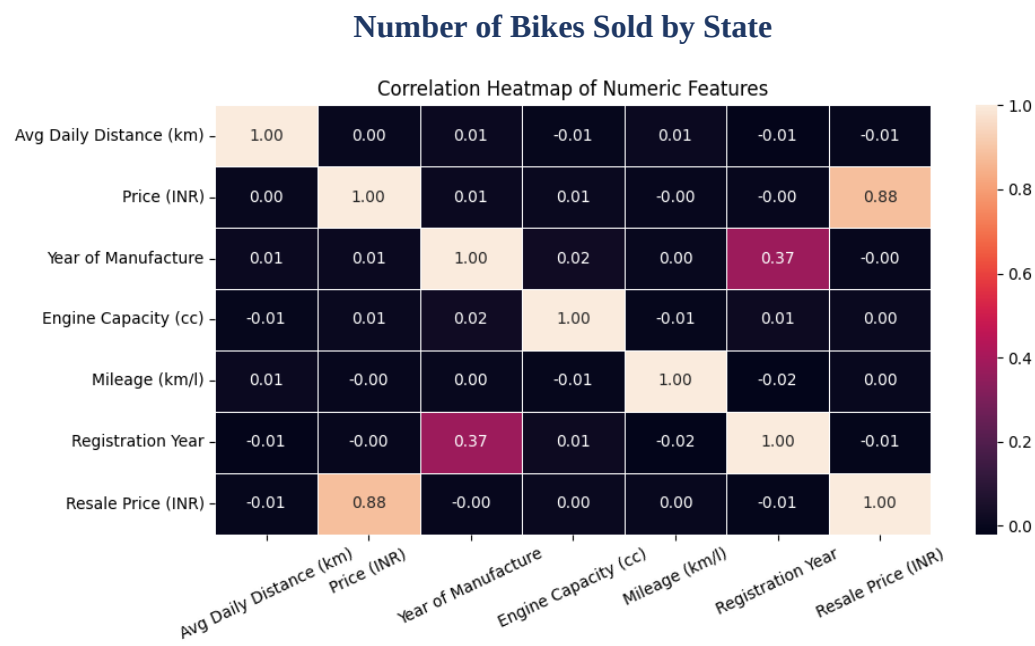
Chart: Scatter Plot

Original Price vs Resale Price (Depreciation Pattern)



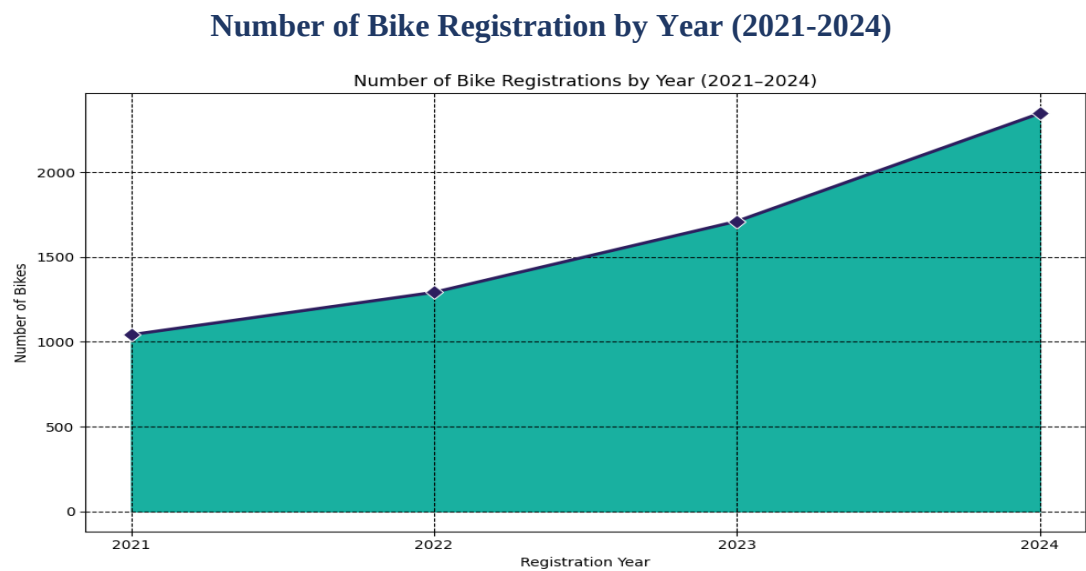
- **Interpretation:** Scatter plot is used to understand the relationship between the original price and resale price. Price (INR) is on the X-axis and Resale Price (INR) on the Y-axis, with Fuel Type used as hue and viridis palette.
 - The chart shows a positive relationship: as the original price increases, the resale price also generally increases.
 - The points are grouped by Hybrid, Electric, and Petrol, and all three fuel types show a similar overall pattern.
-

Chart: Heatmap



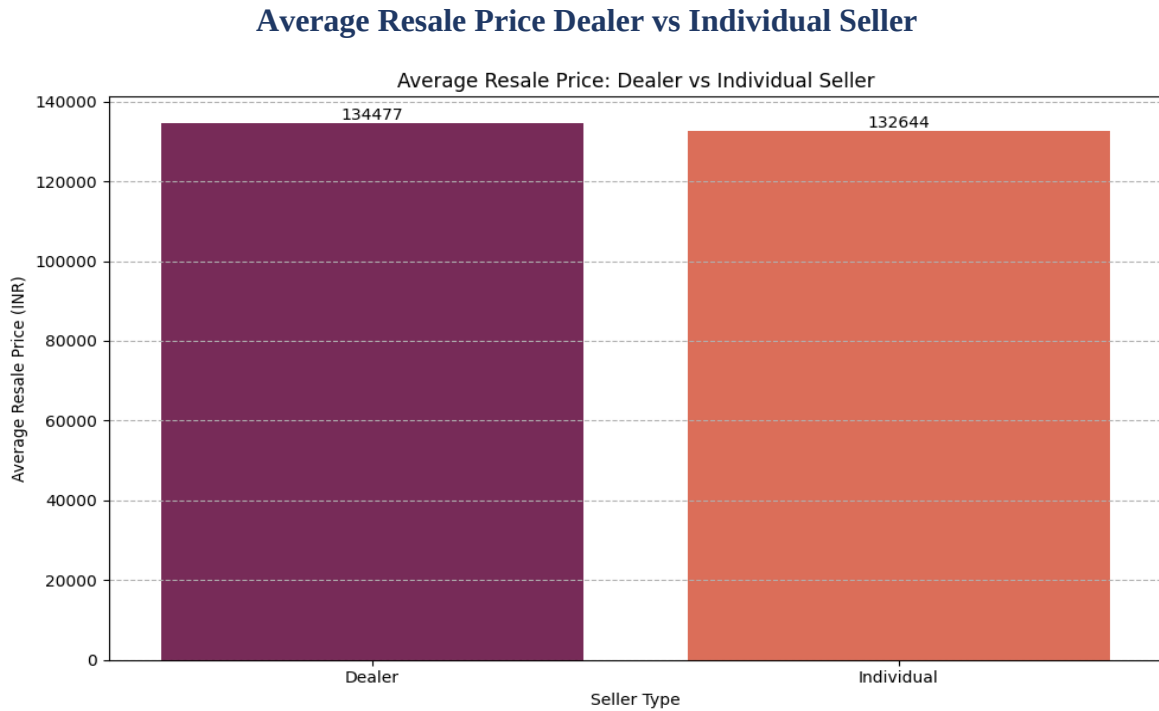
- **Interpretation:** Heatmap is used to understand the correlation between the numerical features. `annot=True`, `fmt=".2f"` and `linewidths=0.5` are used to show the correlation values clearly.
- Price (INR) and Resale Price (INR) have a strong positive correlation of 0.88, showing that higher original price is generally associated with higher resale price.
- Year of Manufacture and Registration Year have a moderate positive correlation of 0.37, while most other features show very low correlation.

Chart: Line Plot



- **Interpretation:** Line plot is used to clearly show the registration trend over the years. Markers, line width, and filled area are used to make the trend easy to understand.
- Bike registrations show a continuous increase from 2021 to 2024.
- 2024 has the highest number of registrations, while 2021 has the lowest, showing a clear upward trend in bike registrations.

Chart: Bar Plot



- **Interpretation:** Bar plot is used to easily compare the average resale price between Dealer and Individual sellers. rocket palette, value labels, and horizontal grid lines are used for better readability.
- Dealer has a slightly higher average resale price (₹1,34,477) compared with Individual sellers (₹1,32,644).
- The difference is small, so the average resale prices of both seller types are quite similar.

8. Analysis and Visualizations (Power BI)

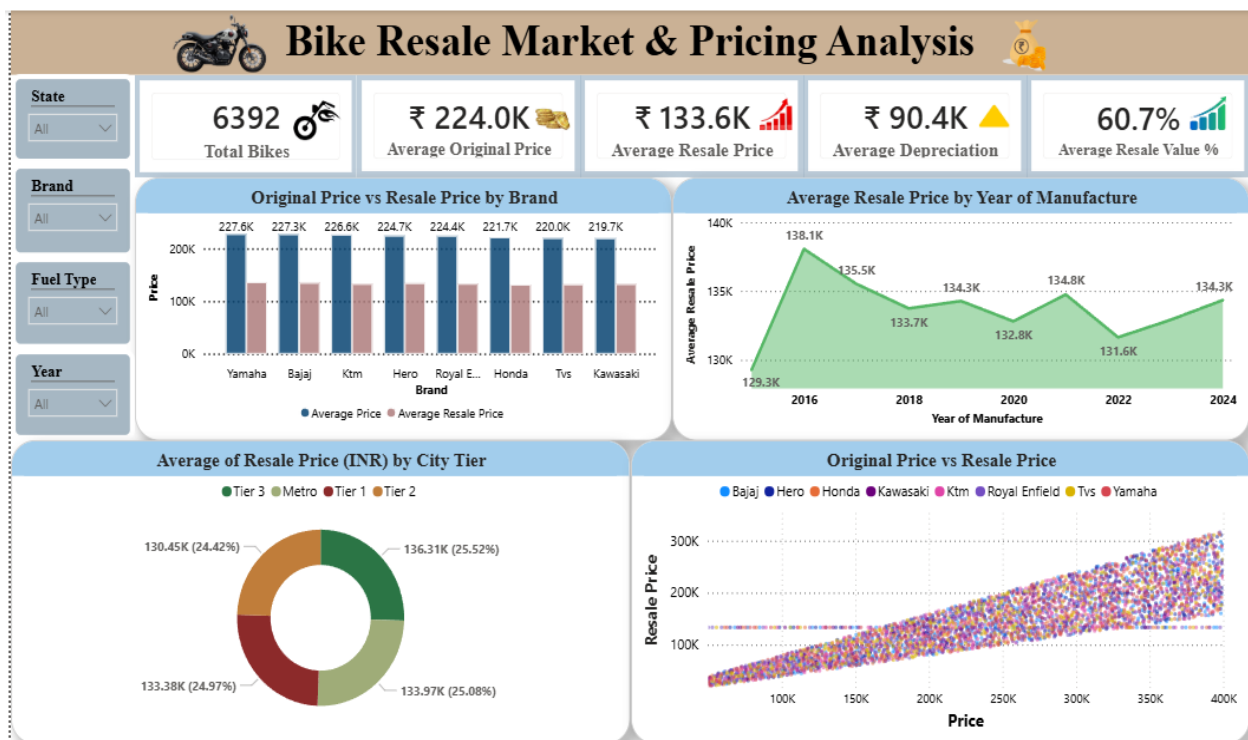
Dashboard Features:

- **Multiple Visualizations:** Used Cards, Bar Charts, Column Charts, Line/Area Charts, Scatter Charts and Donut Charts to present the bike resale analysis.

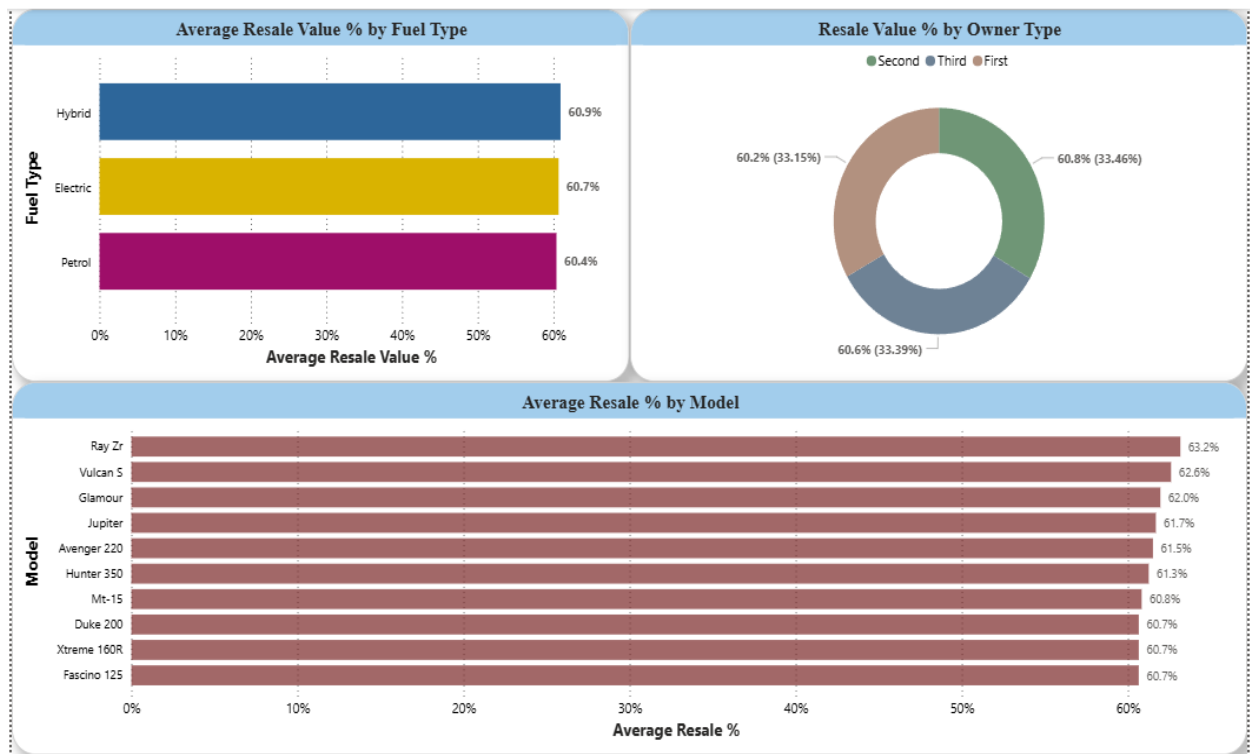
- **Key Performance Cards:** Displayed Total Bikes, Average Original Price, Average Resale Price, Average Resale Value % and Average Depreciation.
- **Comparative Analysis:** Used charts to compare Original Price vs Resale Price, Fuel Type, Owner Type, Brand, Seller Type and Insurance Status.
- **Trend Analysis:** Used year-wise visualizations to understand changes in resale price and resale value percentage.
- **Interactive Report:** Power BI visual interactions and filters are used to explore the data and understand different categories.
- **Clear Titles and Labels:** Each visual has a clear title and suitable labels for easy understanding.
- **Consolidated Dashboard:** Created **two Power BI dashboard pages** to present the overall bike resale market analysis.

SCREENSHOT OF DASHBOARD /REPORT

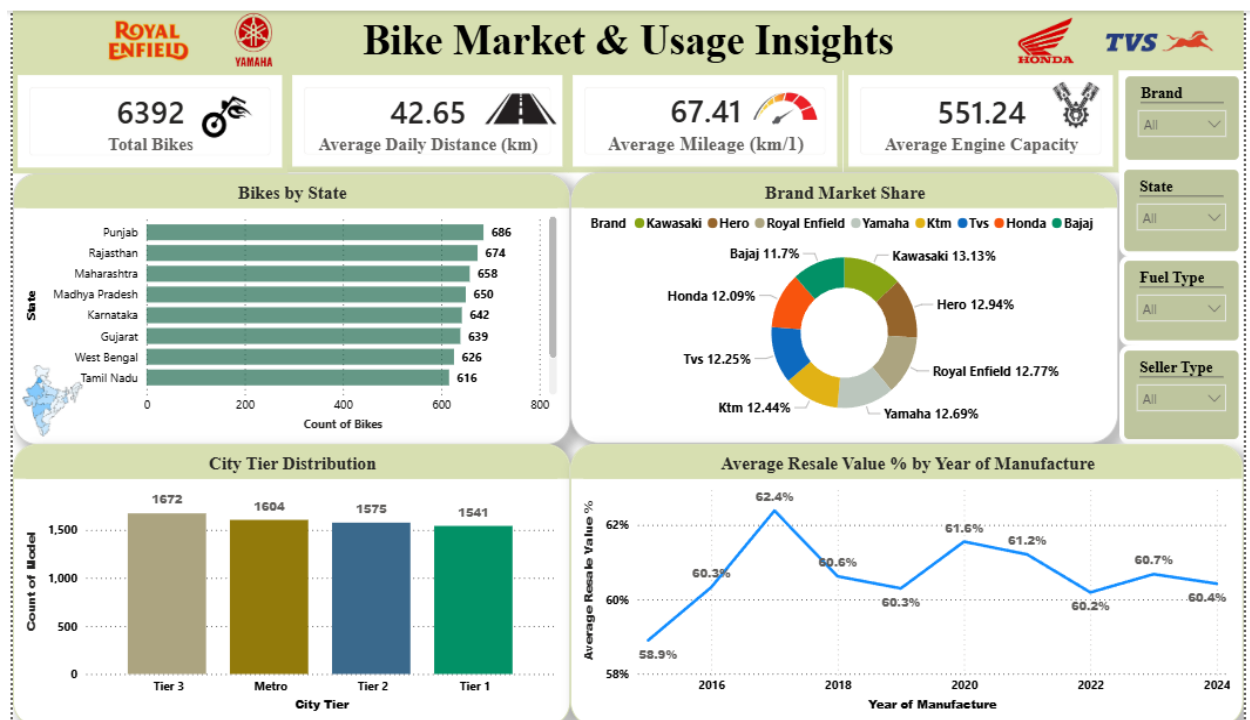
Dashboard 1 (Page-1)



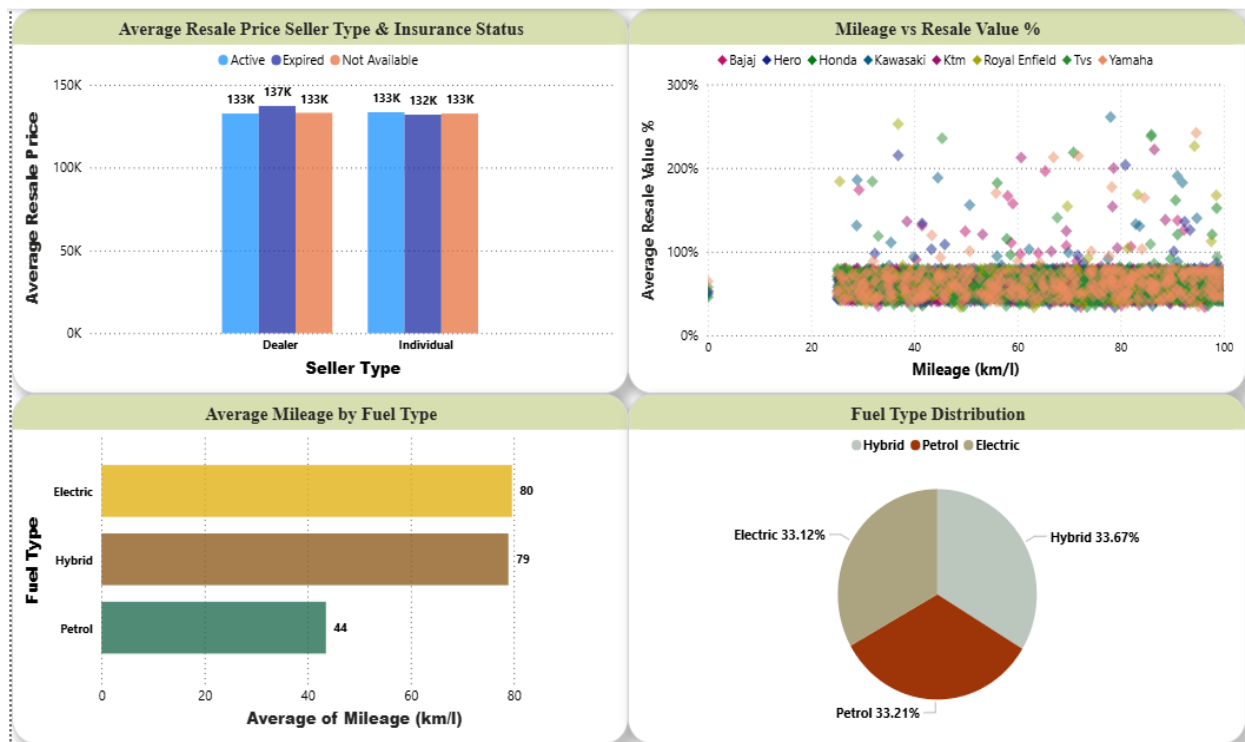
Dashboard 1 (Page-2)



Dashboard 2 (Page-1)



Dashboard 2 (Page-2)



9. Insights & Conclusions

Key Findings: The analysis helped to identified important trends and patterns in the Indian bike resale market based on different factors.

Analysis Insights:

Descriptive

1.State-wise Bike Distribution: Punjab has the highest number of bikes in the dataset with 686 records, followed by Rajasthan with 674 records. This shows that Punjab and Rajasthan have a higher number of bike records compared with the other states in the dataset.

2.Brand Market Share: Kawasaki has the highest brand market share at 13.13%, followed by Hero at 12.94%. The market share of the different brands is quite close, which shows that no single brand has a very large share in the dataset.

3.Resale Value by Fuel Type: Hybrid bikes have the highest average resale value at 60.9%, followed by Electric bikes at 60.7% and Petrol bikes at 60.4%. The difference is small, but Hybrid bikes show slightly better resale value retention.

4.Model-wise Resale Value: Ray ZR has the highest average resale value percentage at 63.2%, followed by Vulcan S at 62.6%. This shows that these models retain a higher percentage of their original price compared with the other models shown in the analysis.

Diagnostic

5.Original Price and Resale Price: The average original price is around 224.0K, while the average resale price is around 133.6K. This shows a clear difference between the original and resale prices and indicates depreciation in bike value.

6.Manufacturing Year and Resale Value: The resale value percentage changes across different manufacturing years. Bikes manufactured in 2017 have the highest resale value percentage at 62.4%, while 2015 has the lowest at 58.9%. This shows that manufacturing year has some effect on resale value retention.

7.Fuel Type and Mileage: Electric bikes have the highest average mileage at 80 km/l, followed by Hybrid bikes at 79 km/l. Petrol bikes have a lower average mileage of 44 km/l in the dataset. This shows a clear difference in mileage between the fuel types.

8.Owner Type and Resale Value: Second-owner bikes have the highest average resale value at 60.8%, followed by First-owner bikes at 60.6% and Third-owner bikes at 60.2%. The difference is small, but the analysis shows a slightly higher resale value for second-owner bikes.

9.Seller Type and Insurance Status: Dealer and Individual sellers have very similar average resale prices across different insurance statuses. The average resale price is around 132K to 137K, showing that seller type and insurance status have only a small difference in average resale price in this dataset.

10. Conclusions

- ❖ This project analysed the Indian two-wheeler resale market using Python and Power BI. The analysis focused on resale price and different factors such as brand, model, fuel type, state, manufacturing year, owner type, seller type and insurance status. The results helped to understand the major trends and differences in bike resale value.
- ❖ The analysis shows that bike model, fuel type and manufacturing year have some differences in resale value, while owner type and seller type show only small differences. The Power BI dashboards helped to present these findings clearly using different charts, cards and visualizations.
- ❖ Overall, this project helped to understand the resale market and depreciation pattern of bikes and can be useful for buyers, sellers and dealers when comparing bike prices and resale values.

Thank You !